PPP-D-711E
December 16, 1975
SUPERSEDING
Fed. Spec. PPP-D-711D
June 23, 1965 and
Fed. Spec. PPP-D-729D
(In part) (see 6.3)

FEDERAL SPECIFICATION

DRUM: METAL, SHIPPING, STEEL, LIGHTWEIGHT (55 GALLON)

This specification was approved by the Commissioner, Federal Supply Service, General Services Administration, for the use of all Federal agencies.

1. SCOPE AND CLASSIFICATION

1.1 Scope. This specification covers new straight side steel drums to be furnished for packing solidified products and powdered or granular (non-regulatory) materials not subject to DOT regulations.

1.2 Classification.

1.2.1 Types. Drums shall be of the following types, as specified (see 6.2):

Type I - Tight head - central cap
Type II - Tight head - offset cap
Type III - Full removable head (cover)
Type IV - Full open head (without cover)

1.2.2 Gage. Drum components shall be of the gage specified (see table I and 6.2).

2. APPLICABLE DOCUMENTS

2.1 The following documents, of the issue in effect on date of invitation for bids or request for proposal, form a part of this specification to the extent specified herein:

Federal Specification:

VV-L-800 - Lubricating Oil, General Purpose, Preservative, (Water Displacing, Low Temperature)

Federal Standards:

FED-STD-101 - Preservatives, Packaging and Packing Materials: Test Procedures
FED-STD-123 - Marking for Shipment (Civil Agencies)

FSC 8110

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(Activities outside the Federal Government may obtain copies of Federal Specifications, Standards, and Handbooks as outlined under General Information in the Index of Federal Specifications and Standards and at the prices indicated in the Index. The Index, which includes cumulative monthly supplements as issued, is for sale on a subscription basis by the Superintendent of Documents, U.S. Government Printing Office, Washington, DC 20402.

(Single copies of this specification and other Federal specifications required by activities outside the Federal Government for bidding purposes are available without charge from Business Service Centers at the General Services Administration Regional Offices in Boston, New York, Washington, DC, Atlanta, Chicago, Kansas City, MO, Fort Worth, Denver, San Francisco, Los Angeles, and Seattle, WA.

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Military Specification:

MIL-G-432 - Gaskets, Synthetic Rubber (for Fuel and Lubricant Containers and Accessories)

Military Standards:

MIL-STD-105 - Sampling Procedures and Tables for Inspection by Attributes
MIL-STD-129 - Marking for Shipment and Storage

(Copies of Military Specficiations and Standards required by suppliers in connection with specific procurement functions should be obtained from the procuring activity or as directed by the contracting officer.)

2.2 Other publications. The following documents form a part of this specification to the extent specified herein. Unless a specific issue is identified, the issue in effect on date of invitation for bids or request for proposal shall apply:

National Motor Freight Traffic Association, Inc., Agent

National Motor Freight Classification

(Application for copies should be addressed to the American Trucking Associations, Inc., Tariff Order Section, 1616 P Street, N.W., Washington, DC 20036.)

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Uniform Clas cation Committee, Agent

Uniform F: ight Classification

(Application for copies should be addressed to the Uniform Classification Committee, Room 1106, 222 South Riverside Plaza, Chicago, IL 60606.)

3. REQUIREMENTS

3.1 Material.

3.1.1 Sheet steel. Sheets for bodies, heads, covers, and caps shall be of the required U.S. Manufacturer's Standard gage, low-carbon, hot or cold-rolled steel of commercial quality. Individual sheets shall conform to table I.

TABLE I. Thickness of sheets

Type	Gage number	Nominal thickness (inch) 1/	Minimum thickness (inch) 1/
III	18	0.0478	0.0428
III 2/	20	0.0359	0.0324
III	22	0.0299	0.0269
I, II, III, IV	24	0.0239	0.0209
I, II, III, IV	26	0.0179	0.0159
I, II, III, IV	28	0.0149	0.0129
I, II, III, IV	30	0.0120	0.0110

1/ Thickness shall be measured at any point on the sheet not less than 3/8 inch from an edge.

2/ 20 gage authorized, when specified (see 6.2).

3.2 Physical requirements. The drums shall be new and shall conform to requirements of the applicable Uniform Freight Classification or National Motor Freight Classification and as hereinafter specified.

3.3 Design. The drum shall be made by curving a flat sheet for the body and either welding or locking and compounding the longitudinal body seam. The welded body seam shall be continuous and free from pinholes (see 4.4). The top (types I and II) and all bottom heads shall be attached to the body by double seaming. Two rolling hoops, two beads or two-series of corrugations shall be rolled or swedged in the body of each drum. The hoops, curl, false wire, beads or corrugations shall conform to good commercial practice.

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3.3.1 Dimensions. The overall height of the drums shall be 35-1/4 inches with a tolerance of plus or minus 1/2 inch. The inside diameter of the drum shall be 22-1/4 inches with a tolerance of plus or minus 3/8 inch.

3.3.2 Tightness. The 18 and 20 gage drums shall not leak when subjected to an internal air pressure of 7 (psi) pounds per square inch and tested as specified in 4.4.2. Drums of 22 to 30 gage shall be light tested as specified in 4.4.1.

3.3.3 Capacity. The rated capacity of the drums shall be 55 gallons. The minimum actual capacity of each drum shall be not less than the marked capacity plus 2 percent. The maximum actual capacity shall be not greater than the marked capacity, plus 5 percent when tested as specified in 4.4.3.

3.3.4 Top head. The top head shall be recessed and types I and II provided with a friction type cover. The edges of the opening for the friction cover shall be drawn inward or upward to prevent sagging and for the protection of handlers. If drawn upward, top edges shall be curled.

3.3.4.1 Type I. Type I drums shall be provided with an opening of 3-1/4 inch minimum diameter and 15 inch maximum diameter, centered in top head and provided with cover to fit, as specified (see 6.2). Caps for curled openings must engage profile of curl.

3.3.4.2 Type II. Type II drums shall be the same as type I drums except with opening located off-center in top head (see 6.2).

3.3.4.3 Types III and IV. The open end of the type III and IV drums shall be provided with a false wire or curl. The cover for the type III drums shall be recessed and fastened to the drum with a gasket and locking ring.

3.4 Gasket. Type III drums shall have a new tubular synthetic rubber, flowed in, or sponge rubber gasket in accordance with MIL-G-432. Gasket material shall be of sufficient thickness and resiliency to provide a seal that will not leak when tested in accordance with 4.4.2.

3.5 Locking ring. The cover for the type III drums shall be secured to the false wire or curl by means of a bolted or lever type locking ring. The locking ring for 18 gage drums shall be not less than 16 gage, and not less than 24 gage for 20 to 30 gage drums as specified (see 6.2).

3.6 Finish.

3.6.1 Preparation of surface. Metal splinters on the interior and exterior of the side seam shall be removed before the heads and bodies are assembled.

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3.6.2 Interior. When specified (see 6.2), the interior of each drum and cover shall be coated with a thin film (use approximately 25 cc's.) of light preservative oil conforming to VV-L-800. Drums to be used for powdered or granulated products shall not be fog sprayed. When specified, (see 6.2), drums to be used for asphalt or tar shall have a release type lining such as clay or other suitable material.

3.6.3 Exterior. The entire exterior of each drum shall be coated with rust-inhibiting and weather-resistant black enamel.

3.7 Marking. Each drum shall be plainly and permanently marked by embossing on the drum bottom as follows:

a. The mark US to indicate that the drum has been procured by the Government. When the drum is procured incident to the product contained therein, the embossing US may be eliminated.

b. The name, initials, or trademark of the manufacturer.

c. The standard gage of steel, the rated capacity of the drum in gallons, and the year of manufacture, shall be abbreviated and appear in the following order: Gage, capacity, year, for example: 24-55-75 indicates that the drum is of 24-gage steel, is of 55-gallons rated capacity, and was made in the year 1975.

d. The size of embossed letters and figures shall be not less than 3/4 inch in height.

3.8 Workmanship. Each finished drum shall be clean, free of rust, slag, foreign matter, and dents. There shall be no nicks, creases or malformed areas on the rolling hoops. Welds shall be free of burns, splatter cracks or incomplete fusion. Spotwelds, if used, shall be free from flash or burn through. Solder shall not be used to repair leaks. Coatings on painted surfaces shall be adherent, free of blisters, imbedded foreign matter, wrinkles, rust, or crazing.

4. QUALITY ASSURANCE PROVISIONS

4.1 Responsibility for inspection. Unless otherwise specified in the contract or purchase order, the supplier is responsible for the performance of all inspection requirements as specified herein. Except as otherwise specified in the contract or order, the supplier may use his own or any other facilities suitable for the performance of the inspection requirements specified herein, unless disapproved by the Government. The Government reserves the right to perform any of the inspections set forth in the specification where such inspections are deemed necessary to assure supplies and services conform to prescribed requirements.

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4.1.1 Inspection. Sampling for inspection shall be performed in accordance with the provisions set forth in MIL-STD-105, except where otherwise indicated.

4.2 Inspection of components and materials. In accordance with 4.1, components and materials shall be inspected in accordance with the requirements of referenced specifications and standards unless otherwise excluded, amended, modified or qualified in this specification or applicable purchase document.

4.2.1 Examination of components and materials. Examination shall be in accordance with table II. Failure to meet any requirement shall be cause for rejection of the component or material.

TABLE II. Examination of components and materials

Examine	Sample	Requirements paragraph
Low carbon steel:		
a. Thickness	1 sheet (each gauge)	3.1
b. Material 1/		3.1
Gasket:		
a. Thickness	5 gaskets	3.4
b. Material 1/		3.4

1/ Material identification shall be made by checking invoices and/or bills of lading.

4.3 Inspection of the end item.

4.3.1 Examination of the end item shall be in accordance with the classification of defects set forth in tables III and IV and the inspection levels and acceptable quality levels (AQLs) specified in 4.3.1.3. The sample unit for this examination shall be one completely fabricated and finished drum. The lot size for purposes of determining the sample size shall be expressed in units of drums.

4.3.1.1 Examination of the end item for defects in construction and workmanship.

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TABLE III. Examination of the end item for defects in construction and workmanship.

Examine	Defect
Construction:	
Design	Longitudinal body seam not as specified. Attached top and bottom heads not double seamed. Less than 2 rolling hoops, beads, or series of corrugations on body of drum. Nicks, creases, or malformed areas on rolling hoops.
Top head	Not recessed as specified. Types I and II not provided with friction cover as specified. Edges of opening for friction cover not drawn inward or not curled when drawn upward.
Types I and II	Openings and covers not size specified.
Type III	Cover not fastened with ring as specified. Cover does not fit over false wire. Gasket missing or damaged or not type specified. Locking ring not as specified or does not fit. Jam and nut missing. Wrong size bolt.
Finish	Metal splinters on side seam. Interior of drums not as specified. Not completely fogged sprayed. Exterior not coated as specified. Evidence of dirt or rust, blisters, wrinkles and crazing.
Sideseam	Sideseam not continuous. Evidence of pinholes.

4.3.1.2 Examination of the end item for dimensional and marking defects.

TABLE IV. Examination of the end item for dimensional and marking defects.

Examine	Defect
Dimensions	Overall height of drum more or than 35-3/4 inches or less than 34-3/4 inches. Inside diameter more than 22-5/8 inches or less than 21-7/8 inches.

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TABLE IV. Examination of the end item for dimensional and marking defects (cont'd).

Examine	Defect
Type I and II drum	Cap dimension not friction fit.
Capacity	Nominal capacity more or less than specified. Actual capacity not as specified.
Marking	Embossing not on bottom head as specified. Omitted, incomplete, incorrect, illegible, improper size, location or sequence or method of application.

4.3.1.3 Inspection levels and acceptable quality levels (AQLs) or examination. The inspection levels for determining the sample size and the acceptable quality levels expressed in defects per 100 units shall be as follows:

Table No.	Inspection level	AQL
III	S-3	6.5
IV	S-2	2.5

4.3.2 Examination of preparation for delivery. The marking shall be examined for conformance to the requirements of section 5.

4.4 Tests.

4.4.1 Seam test. A light shall be inserted in each drum, selected in accordance with 3.3 and 3.3.2 to verify that the side seam is continuous and free from pinholes. Defective drums shall be rejected or repaired and retested.

4.4.2 Leak test. Drums shall be subjected to an internal air pressure of 7 psi, seams submerged in water, or covered with soapsuds or heavy oil, and then checked for leaks to determine compliance with 3.3.2 and 3.4.

4.4.3 Capacity test. The capacity of the drum shall be determined for conformance with 3.3.3 in accordance with Method 5010 of FED-STD-101.

5. PREPARATION FOR DELIVERY

5.1 Packing. The drums shall be prepared for shipment in a manner to insure carrier acceptance and safe delivery at destination at lowest transportation rate for such supplies. Shipments shall be in accordance with the Uniform Freight Classification or National Motor Freight Classification, as applicable.

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5.2 Marking for direct shipment to the Government.

5.2.1 Civil agencies. In addition to any special marking required by the contract or order, shipments shall be marked in accordance with the requirements of FED-STD-123.

5.2.2 Military requirements. In addition to any marking required by the contract or order, shipments shall be marked in accordance with the requirements of MIL-STD-129.

6. NOTES

6.1 Intended use. Drums covered by this specification are intended for use by the Government, or for the shipment of commodities purchased by the Government, when prescribed in the applicable commodity specification. Commodities shall consist of solidified products and powdered or granular (nonregulatory) materials not subject to DOT regulations.

6.2 Ordering data. Purchasers should exercise any desired options offered herein and procurement documents should specify the following:

(a) Title, number, and date of this specification.
(b) Type required (see 1.2.1).
(c) Gage (see 1.2.2 and table I).
(d) Size of opening required (see 3.3.4.1 and 3.3.4.2).
(e) Locking ring bolted or lever type (see 3.5).
(f) When the interior of drum shall be fog sprayed (see 3.6.2).
(g) When interior of drum shall have a release type lining (see 3.6.2).
(h) Shipment markings required (see 5.2).

6.3 Supersession data. This specification includes as type III the requirements for type III drums in PPP-D-729 dated September 15, 1972.

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